

2.6 Propositional Equivalence

Propositional equivalence explains when two logical expressions have the same meaning. Even if expressions look different, they may give the same result.

- Propositional equivalence is checked using truth tables.
- If two expressions have the same truth values in all cases, they are equivalent.

Symbol Used for Propositional equivalence

- The symbol “ $\equiv$ ” or $\Leftrightarrow$ is used to show equivalence.

Purpose of Propositional equivalence

- Propositional equivalence improves clarity in reasoning.

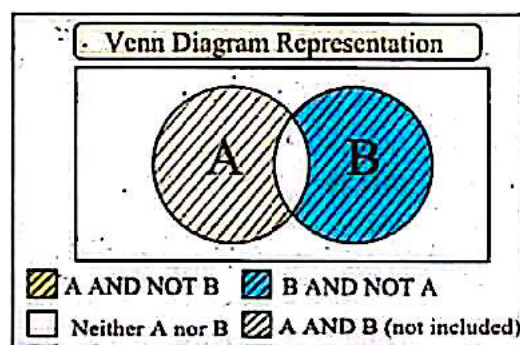
Expression 1: $(A \text{ AND NOT } B) \text{ OR } (B \text{ AND NOT } A)$

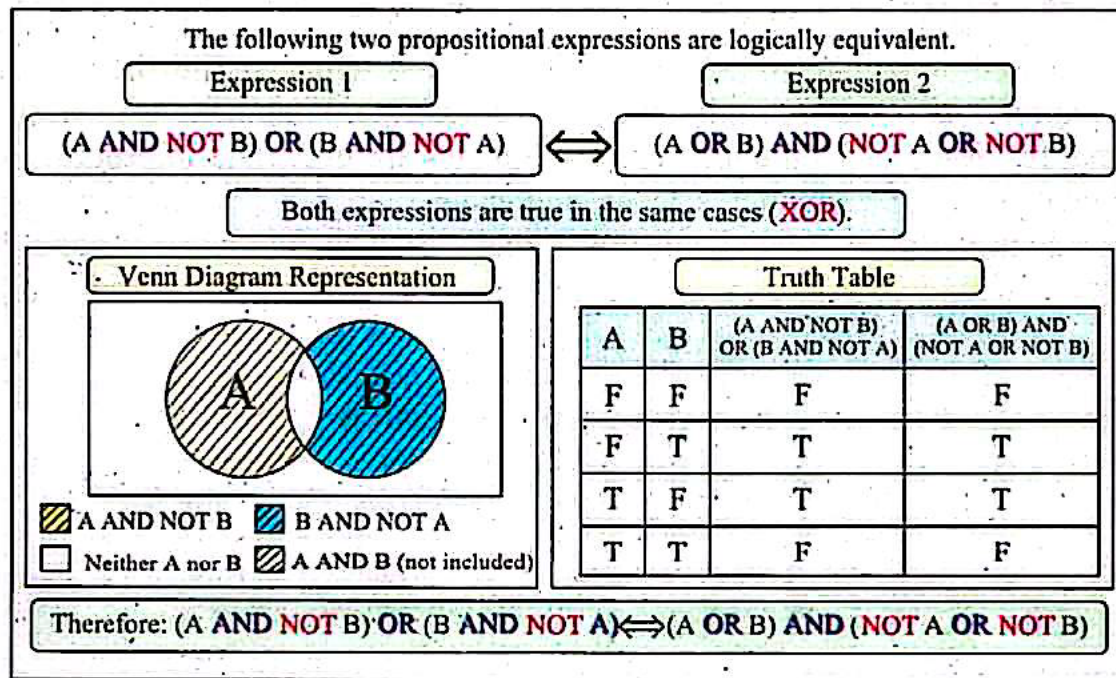
A	B	NOT B	NOT A	A AND NOT B	B AND NOT A	(A AND NOT B) OR (B AND NOT A)
False	False	True	True	False	False	False
False	True	False	True	False	True	True
True	False	True	False	True	False	True
True	True	False	False	False	False	False

Expression 2: $(A \text{ OR } B) \text{ AND } (\text{NOT } A \text{ OR } \text{NOT } B)$

A	B	A OR B	NOT B	NOT A	NOT A OR NOT B	(A OR B) AND (NOT A OR NOT B)
False	False	False	True	True	True	False
False	True	True	False	True	True	True
True	False	True	True	False	True	True
True	True	True	False	False	False	False

Expression 1: $(A \text{ AND NOT } B) \text{ OR } (B \text{ AND NOT } A)$





TIDBIT Equivalent expressions may look different but always give the same result; they help simplify complex logic.

Identifying Equivalent Logical Expressions

- Test each expression for all possible truth values using a truth table. If the results match in every case, the expressions are equivalent.
- Logical laws can also help identify equivalence and simplify expressions.

Importance of Equivalence in Logic

- Helps simplify logic problems
- Allows replacing complex expressions with simpler ones
- Saves time and effort
- Improves code efficiency in programming
- Helps avoid logical errors
- Ensures correct reasoning and decisions
- Useful in mathematics and computer science

Short Questions

1. What is propositional equivalence?
2. How can we check whether two logical expressions are equivalent?
3. What symbol is used to show equivalence?
4. Why is propositional equivalence important?
5. How does equivalence help in programming?
6. Can two expressions look different but still be equivalent? Explain.
7. What is the benefit of identifying equivalent expressions?